

AZAAN YASSER YAQUB

Mechanical Engineering Student | Robotics & Computer Vision (CV) | Seeking Summer 2027 Internships

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EDUCATION

Imperial College London

London, UK | Sep 2024 – Jun 2028

Mechanical Engineering (M.Eng.)

Grade: First Class Honours (1:1) as of 2nd Year

Societies: Imperial College AUV (Co-founder), Imperial College Robotics, Imperial College American Football

Jebel Ali School

Dubai, UAE | Aug 2020 – May 2024

A-Level: A*A*A*A (Mathematics, Physics, Economics, Chemistry) | GCSE: 7 9s, 2 8s

SKILLS

- **Computer Vision & Machine Learning (CV/ML):** Python, OpenCV, YOLO Object Detection, MediaPipe, DeepFace, PyTorch, Convolutional Neural Networks (CNNs), Kalman Filters
- **Robotics, Autonomy & Deployment:** ROS 2 (Robot Operating System), Docker, Linux, Edge Deployment, Git
- **Mechanical Design:** SolidWorks (CAD), GD&T, Manual Machining, DFM

WORK EXPERIENCE

PwC | Frontier Labs Intern (*Mechatronics & Computer Vision*)

Dubai, UAE | Jun 2026 – Aug 2026

- Engineered Telebot's perception pipeline - YOLO detection on an NVIDIA Jetson at ~15 FPS driving a PD-controlled visual servo loop - enabling a 6-DOF arm to track and follow a person, with hold-on-fault safety on detection dropout
- Split vision and motor control into isolated Python runtimes and deployed to production via Docker Compose and systemd
- Streamed live video to 10+ concurrent viewers over an authenticated, encrypted WebSocket at sub-30ms latency, with live mode-switching between autonomous tracking and manual teleoperation
- Paired the arm with Meta Ray-Ban smart glasses for a first-person teleoperation demo, showcased to 1,000+ attendees at a week-long Department of Government Enablement event in Abu Dhabi, presented to visiting government officials
- Built VisionHub's dispatch layer, routing frames to 3 isolated ML services (MediaPipe, YOLO26, DeepFace) with self-registering health checks that distinguish a crashed worker from one timing out
- Led VisionHub's facial recognition features using DeepFace, hitting 90% accuracy across 10+ users; then extended it into CCTV attendance tracking with present/absent/indeterminate loggings to avoid misreading degraded footage as absent

Avanza Solutions | AI Research Intern

Dubai, UAE | Jul 2025 – Aug 2025

- Built a Python RAG pipeline with the OpenAI API and ChromaDB for context-aware document retrieval; then layered a chatbot with token streaming and multi-turn memory, supporting 3+ concurrent sessions at ~500ms time-to-first-token
- Benchmarked 5+ open-source LLMs against the OpenAI API across zero-shot, few-shot, and Chain-of-Thought prompting, identifying the strategy that improved accuracy by 13% and informed the team's production model choice

Evan Lim Penta | MEP Engineering Intern

Dubai, UAE | Jun 2025

- Inspected chilled water piping systems on a 12-storey development, checking installation against MEP blueprints and building code - working alongside engineers on a 200+ person site team to catch discrepancies before sign-off

PROJECTS & LEADERSHIP

Imperial College AUV Society | Co-founder, Head of Automation

London, UK | Nov 2025 – Present

- Co-founded Imperial's first Autonomous Underwater Vehicle (AUV) society (5+ members), building toward a 5-vehicle swarm with inter-vehicle communication protocols and synchronised bathymetric surveys to map lakes/rivers
- Leading perception and autonomy: currently prototyping real-time obstacle avoidance (OpenCV, CNN-based tracking) and a ROS 2 motion-control architecture, aiming for a 3m operating depth and 10 – 20cm mapping resolution

StepFusion | Personal Project - [GitHub](#)

London, UK | Sep 2026 – Present

- Built a pedestrian dead-reckoning system using an iPhone's IMU with gyroscope/magnetometer fusion via a Kalman filter; traced a heading-correction failure to magnetic interference and redesigned it, cutting position error 12.7%

EduNet | Personal Project - [GitHub](#)

London, UK | Aug 2026 – Present

- Built a neural network from scratch in NumPy (forward/backprop + gradient descent), packaged as a pip-installable library with swappable activations/cost functions, gradient checking accurate to 1e-7, and a live network visualiser

Imperial Design Transmission Project | Project Manager

London, UK | Sep 2025 – Dec 2025

- Led a 5-member team through a 10-week design-to-manufacture project (3-stage F24 drivetrain), achieving a 9.5:1 gear ratio at 20.1 km/h against a 25 km/h constraint; FMEA identifying chain derailment as the highest-priority failure risk

CERTIFICATIONS & ACHIEVEMENTS

- AI Ventures Hackathon: 3rd of 30 teams (£500 prize) for an ML-driven construction procurement/pricing platform
- AI for Autonomous Vehicles and Robotics | University of Michigan: implemented Kalman filter state estimation in Python; studied SLAM and R-CNN object detection architectures